Agenda

Business Analytics

- Introduction
- Types of Data and Databases
- Data Models
- Entity Relationship Model
- Case 1: Reverse Engineer a Database Model
- The Star and Snowflake Schema
- Analyzing Data with Power BI
- Case 2: Analyzing Procurement Transactions

Data Science

Case 2: Build a BI-report in Power BI

You are tasked with building a Business Intelligence report for the Procurement Department of Rustic Roasters, a coffee trading company. The company is experiencing significant challenges with late purchase orders arriving at their warehouses. The Procurement Department wants to gain deeper insights into this issue and identify potential causes for these delays.

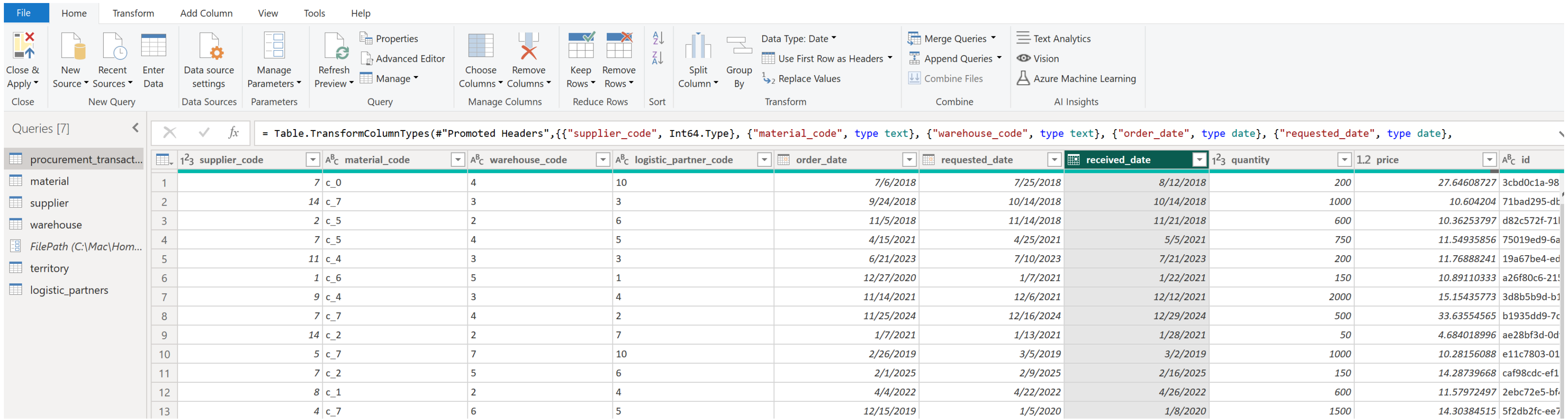
You have been provided with the following data files:

- customer.xlsx
- material.xslx
- procurement_transactions_part_A.xlsx
- procurement_transactions_part_B.xlsx
- supplier.xlsx
- territory.xlsx
- warehouse.xlsx
- logistics_partners.json

Your report should leverage these datasets to analyze the current situation and support data-driven decision making for the Procurement Department.

IMPORTING DATA AND CREATING A DATA MODEL

Read an Xlsx File – Ensure Correct Datatypes



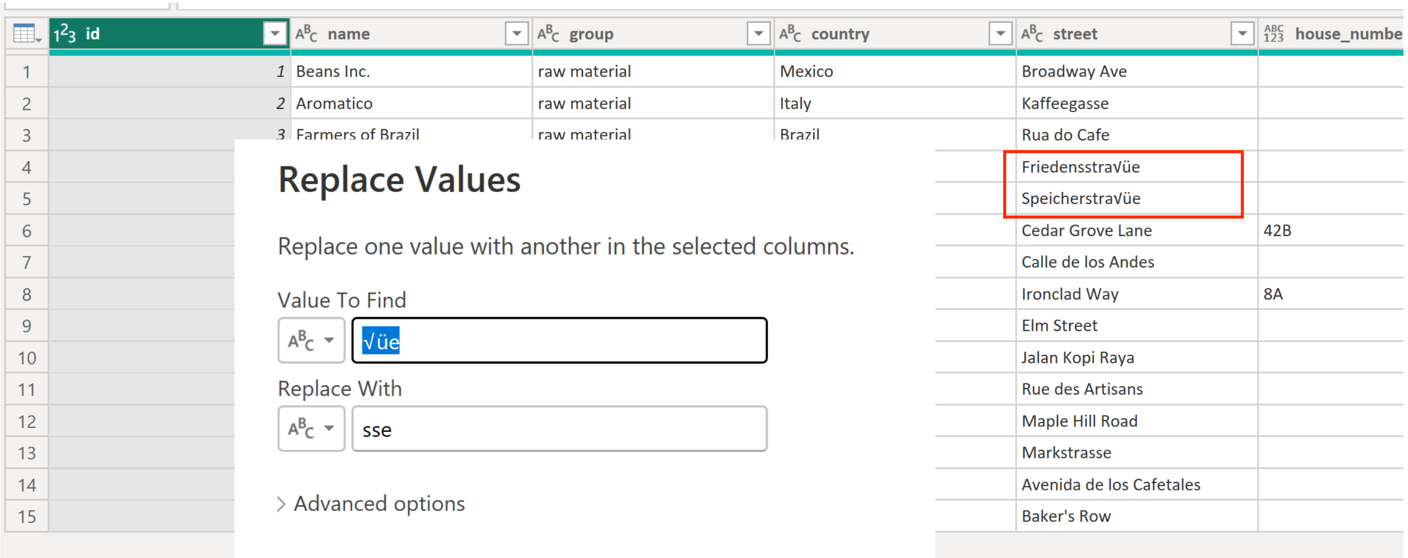
Read an Xlsx File – Ensure Correct Datatypes

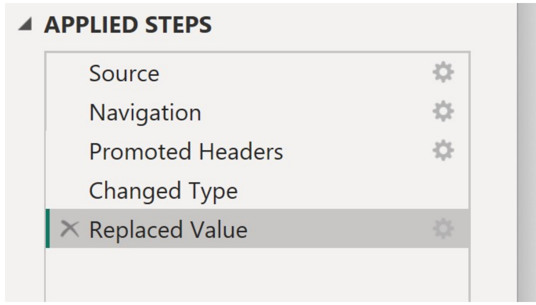
Table	Column	Type
customer	id	Text
procurement_transactions_part_a	supplier_code	Text
procurement_transactions_part_a	warehouse_code	Text
procurement_transactions_part_a	logistics_partner_code	Text
procurement_transactions_part_b	order_date	Date
procurement_transactions_part_b	requested_date	Date
procurement_transactions_part_b	received_date	Date

Read an Xlsx File – Ensure Correct Datatypes

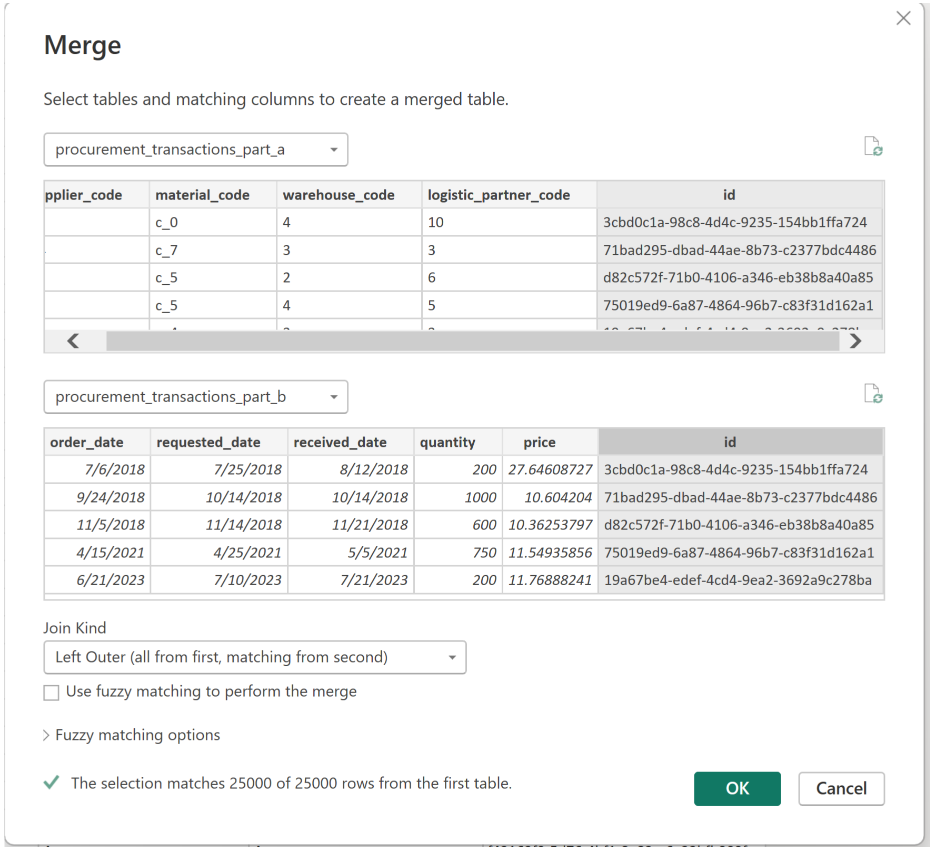
Table	Column	Type
supplier	id	Text
supplier	house_number	Text
Supplier	postal_code	Text
Territory	id	Text
Warehouse	id	Text
Warehouse	house_number	Text
Warehouse	postal_code	Text

Replace "Wrong" Values in Supplier Table – All Steps Are Tracked and Can Be Reversed

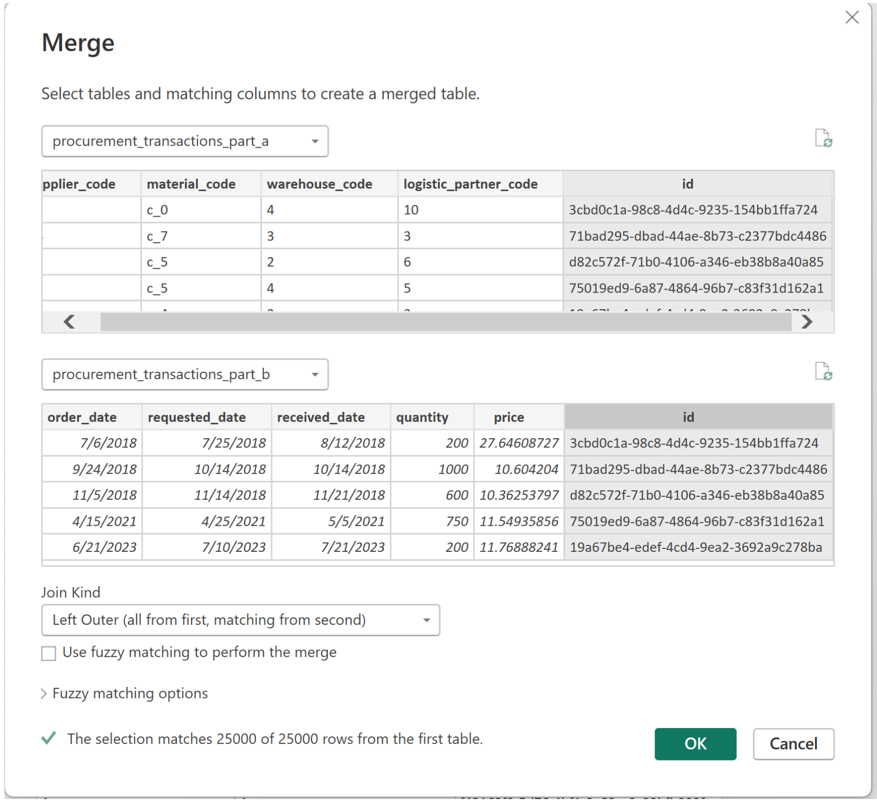




Merge procurement_transactions_part_a and procurement_transactions_part_b as New



Merge procurement_transactions_part_a and procurement_transactions_part_b as New



Merge

Select tables and matching columns to create a merged table.

procurement_transactions_part_a

pplier_code	material_code	warehouse_code	logistic_partner_code	id
	c_0	4	10	3cbd0c1a-98c8-4d4c-9235-154bb1ffa724
	c_7	3	3	71bad295-dbad-44ae-8b73-c2377bdc4486
	c_5	2	6	d82c572f-71b0-4106-a346-eb38b8a40a85
	c_5	4	5	75019ed9-6a87-4864-96b7-c83f31d162a1

procurement_transactions_part_b

order_date	requested_date	received_date	quantity	price	id
7/6/2018	7/25/2018	8/12/2018	200	27.64608727	3cbd0c1a-98c8-4d4c-9235-154bb1ffa724
9/24/2018	10/14/2018	10/14/2018	1000	10.604204	71bad295-dbad-44ae-8b73-c2377bdc4486
11/5/2018	11/14/2018	11/21/2018	600	10.36253797	d82c572f-71b0-4106-a346-eb38b8a40a85
4/15/2021	4/25/2021	5/5/2021	750	11.54935856	75019ed9-6a87-4864-96b7-c83f31d162a1
6/21/2023	7/10/2023	7/21/2023	200	11.76888241	19a67be4-edef-4cd4-9ea2-3692a9c278ba

Join Kind

Left Outer (all from first, matching from second)

Use fuzzy matching to perform the merge

Fuzzy matching options

The selection matches 25000 of 25000 rows from the first table.

OK Cancel

Combining Data

Orders

order_id	customer_id	product	quantity
1	1	Apples	10
2	2	Bananas	20
3	4	Cherries	30
4	1	Dates	40

Customers

id	name	city	email	contact
1	Smith	New York	smit@...	07979
2	Doe	Frankfurt	doe@...	2347
3	Wang	Munich	wang@	143

Combining Data: Left Join - Using All Rows From the First Table

order_id	customer_id	product	quantity	id	name	city	email	contact
1	1	Apples	10	1	Smith	New York	smit@...	07979
2	2	Bananas	20	2	Doe	Frankfurt	doe@...	2347
3	4	Cherries	30	NULL	NULL	NULL	NULL	NULL
4	1	Dates	40	1	Smith	New York	smit@...	07979

Pandas equivalent:

```
pd.merge(customers, suppliers, how='left', left_on='customer_id', right_on='id')
```

Combining Data: Right Join - Using All Rows From the Second Table

order_id	customer_id	product	quantity	id	name	city	email	contact
1	1	Apples	10	1	Smith	New York	smit@...	07979
2	2	Bananas	20	2	Doe	Frankfurt	doe@...	2347
4	1	Dates	40	1	Smith	New York	smit@...	07979
NULL	NULL	NULL	NULL	3	Wang	Munich	wang@	143

Pandas equivalent:

```
pd.merge(customers, suppliers, how=‘right', left_on='customer_id', right_on='id')
```

Combining Data: Outer Join - Using All Rows From Both Tables

order_id	customer_id	product	quantity	id	name	city	email	contact
1	1	Apples	10	1	Smith	New York	smit@...	07979
2	2	Bananas	20	2	Doe	Frankfurt	doe@...	2347
3	4	Cherries	30	NULL	NULL	NULL	NULL	NULL
4	1	Dates	40	1	Smith	New York	smit@...	07979
NULL	NULL	NULL	NULL	3	Wang	Munich	wang@	143

Pandas equivalent:

```
pd.merge(customers, suppliers, how='outer', left_on='customer_id', right_on='id')
```

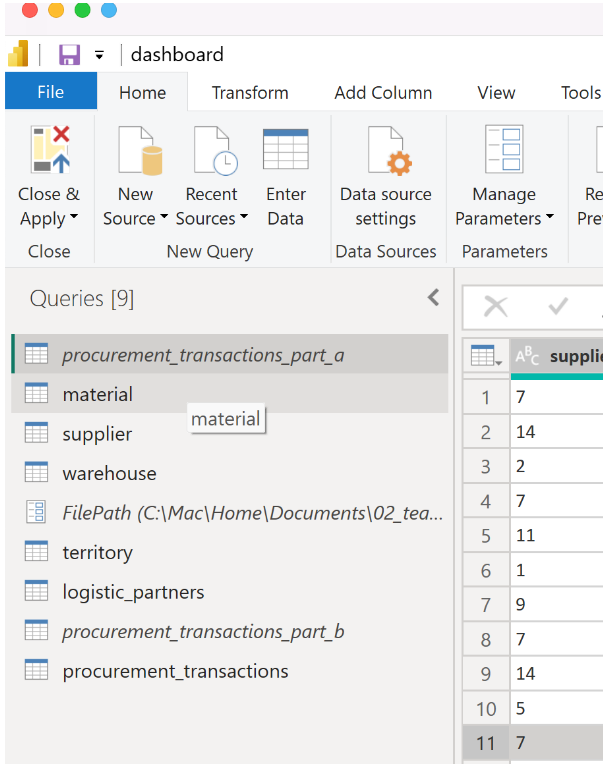
Combining Data: Inner Join - Using Only Rows for Which Data is Found in Both Tables

order_id	customer_id	product	quantity	id	name	city	email	contact
1	1	Apples	10	1	Smith	New York	smit@...	07979
2	2	Bananas	20	2	Doe	Frankfurt	doe@...	2347
4	1	Dates	40	1	Smith	New York	smit@...	07979

Pandas equivalent:

```
pd.merge(customers, suppliers, how='inner', left_on='customer_id', right_on='id')
```

Disable the load for procurement transactions_part_a and …_part_b



Defining the Data Model

Manage relationships

New relationship | Autodetect | Edit | Delete | Filter

From: table (column)	Relationship	To: table (column)	Status
procurement_transactions (log...	* ◀ 1	logistic_partners (id)	Active
procurement_transactions (ma...	* ◀ 1	material (item_code)	Active
procurement_transactions (su...	* ◀ 1	supplier (id)	Active
procurement_transactions (wa...	* ◀ 1	warehouse (id)	Active
supplier (country)	* ◀ 1	territory (name)	Active

CREATING CALCULATED COLUMNS IN DAX

Create a Measure to Calculate Days Late

```
Days Late = DATEDIFF(
                procurement_transactions[requested_date].[Date],
                procurement_transactions[received_date].[Date],
                DAY
                )
Late = IF(
        procurement_transactions[requested_date].[Days Late] > 0,
        1,
        0
        )
```

Example Build Categories of Late Purchases Orders with a Calculated Column Using the IF-Operator

```
Category = IF(
               procurement_transactions[Days Late] < 1, "A",
               IF (
                      procurement_transactions[Days Late] < 5, "B",
                      IF (
                             procurement_transactions[Days Late] < 10, "C",
                             "D"
               )
```

More Efficient Version Using the SWITCH-Operator

```
Category = SWITCH(
        TRUE(),
        procurement_transactions[Days Late] < 1, "A",
        procurement_transactions[Days Late] < 5, "B",
        procurement_transactions[Days Late] < 10, "C",
        "D"
        )
```

Example Using Switch for Getting the Name of the Weekday

```
Requested Weekday = SWITCH (
procurement_transactions[requested_date].[Day],
    1, "Monday",
    2, "Tuesday",
    3, "Wednesday",
    4, "Thursday",
    5, "Friday",
    6, "Saturday",
    7, "Sunday",
    "invalid"
    )
```

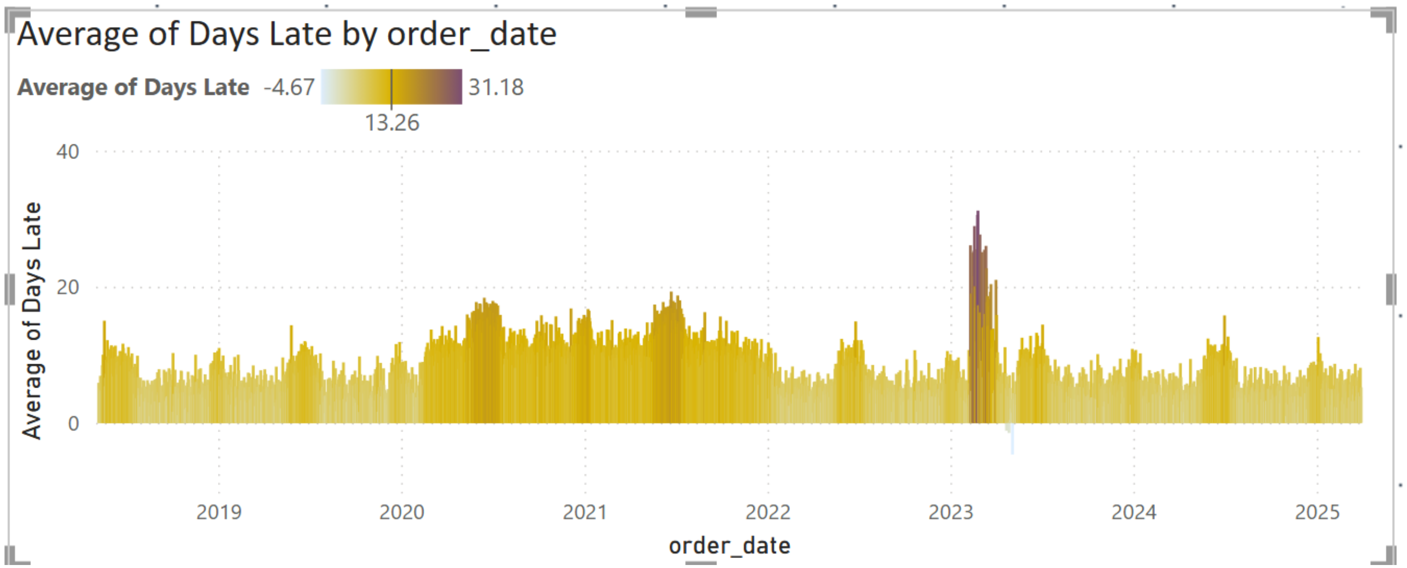
Case 2: Build a BI-report in Power Bi – Task Part 2

- Many companies faced significant transport challenges during the COVID-19 pandemic. Did Rustic Roasters also experience similar issues? Please analyze the data to identify any pandemic-related disruptions.

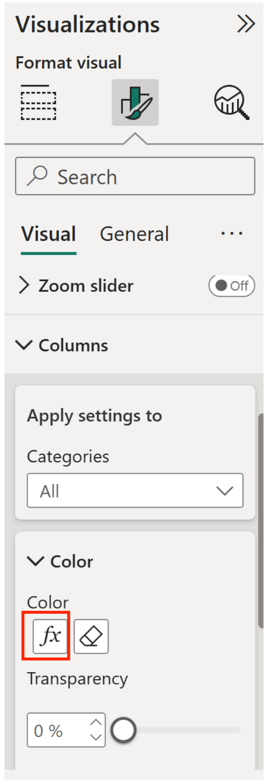
- Additionally, there was a workers' strike in the past. Can you determine which warehouse(s) were affected by the strike?

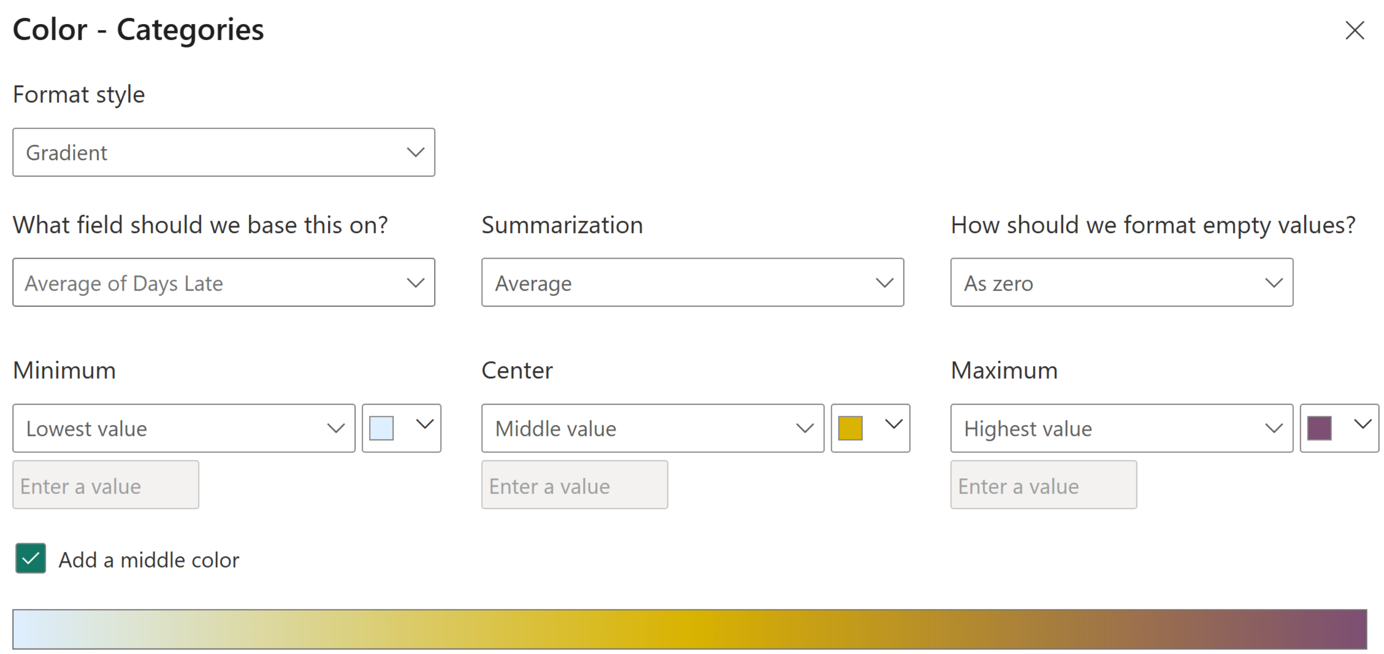
ADVANCED FORMATTING OF VISUALS

Highlighting Effects

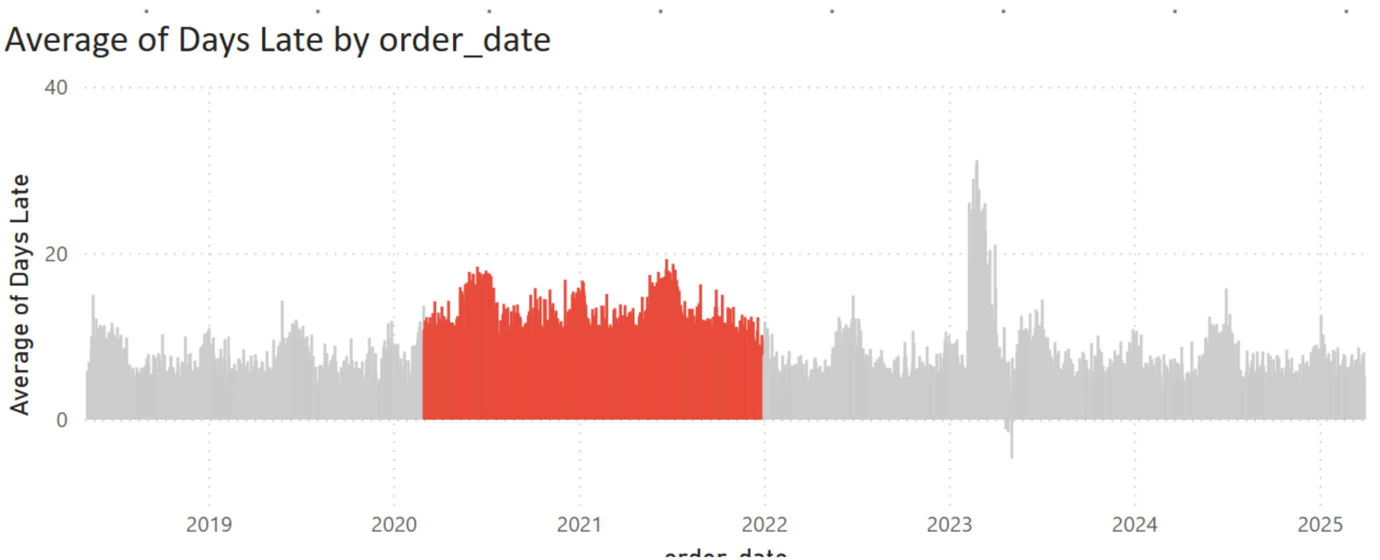


Highlighting Effects





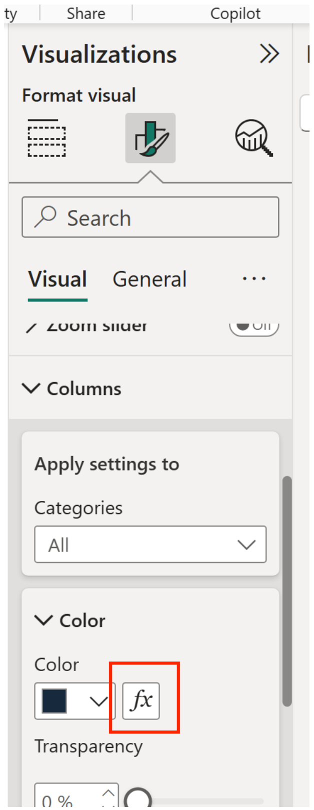
Highlighting Corona Effect



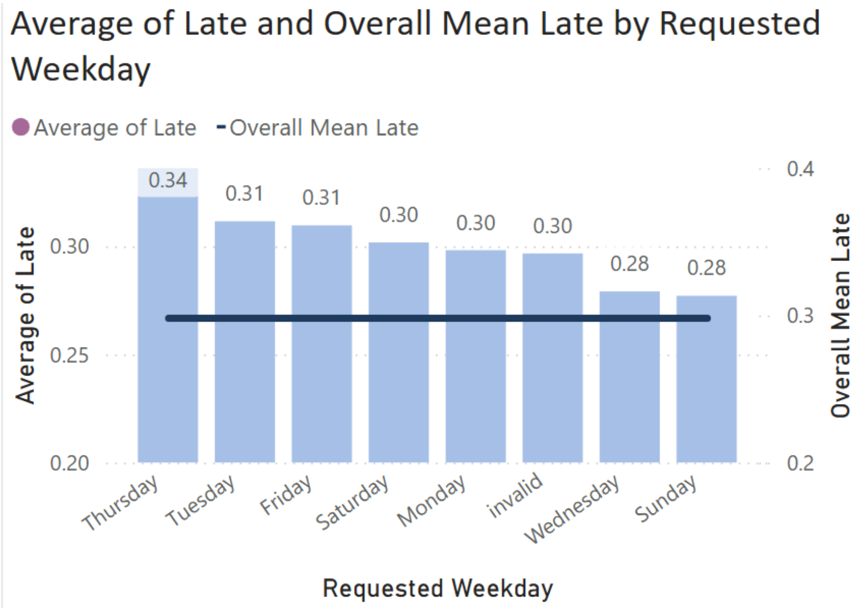
Highlighting Corona Effect: 1. Create a Column for the Color

```
Corona = IF (
        procurement_transactions[order_date] > DATE(2020,3,1) &&
        procurement_transactions[order_date] < DATE(2022, 1, 1),
        "#E74C3C",
        "#CCCCCC”
)
```

Highlighting Corona Effect: Using the Column for Coloring the Bars



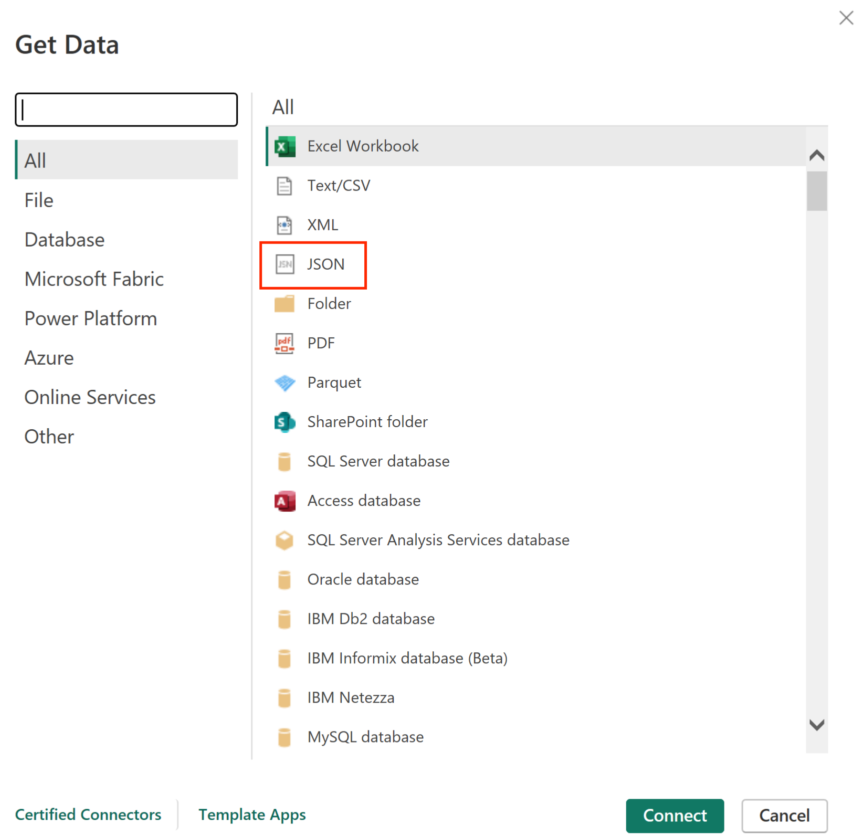
Adding the Mean Value for a Variable Ignoring Categories




```
Overall Mean Late = CALCULATE(
    AVERAGE(procurement_transactions[Late]),
    ALL(procurement_transactions)
)
```

LOADING DATA IN OTHER FORMATS

Loading Java Script Object notation (JSON) data

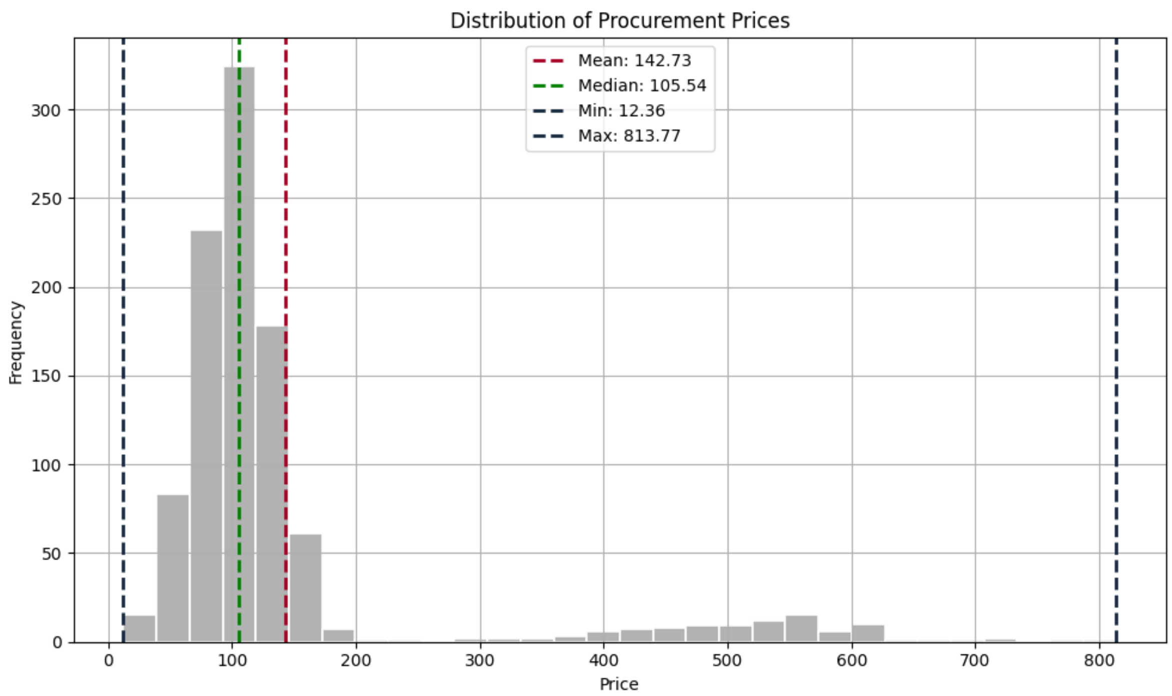


Case 2: Build a BI-report in Power Bi – Task Part 3

- Load the logistic_partners.json file
- Convert the columns to the appropriate data types
- Integrate the data into the data model
- Analyze whether the choice of logistics partner influences whether an order is late.

STATISTICAL MEASURES TO DESCRIBE DATA

Describing Metric Variables



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Describing Metric Variables

Sample Data:
Prices=[2, 4, 7, 10, 15]

	Explanation	Formula	Value
Mean		$mean = \frac{1}{N}\sum x_i$	
Median	The median is the middle value when data is sorted in order. It expresses the point that splits the data in half.	$med = x_{\left(\frac{n+1}{2}\right)}$ *(Odd number of elements)* $med = \frac{x_{\left(\frac{N}{2}\right)} + x_{\left(\frac{N+1}{2}\right)}}{2}$ *(Even number of elements)*	
Variance	The variance measures how spread out the data is around the mean.	$var = \frac{1}{N}\sum (x_i - \bar{x})^2$	

Describing the Relation between to Metric Variables

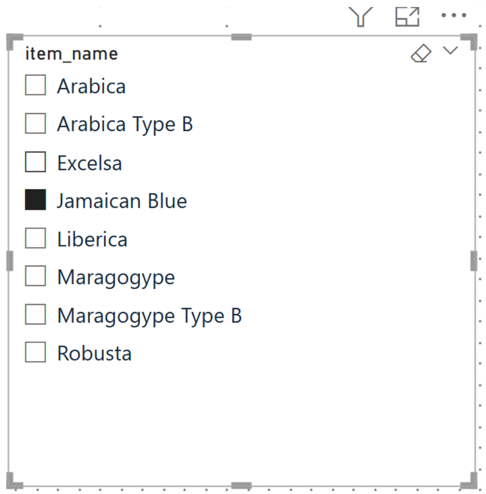
	Explanation	Formula
Covariance	Positive covariance → when one variable increases, the other tends to increase. Negative covariance → when one increases, the other tends to decrease.	$cov = \frac{1}{N}\sum(x_i - mean(x))\big(y_j - mean(y)\big)$
Correlation Coefficient (r)	r=+1: perfect positive linear relationship r = -1: perfect negative linear relationship r=0 : no linear relationship	$r = \frac{Covariance(x,y)}{\sqrt{var(x)var(y)}}$

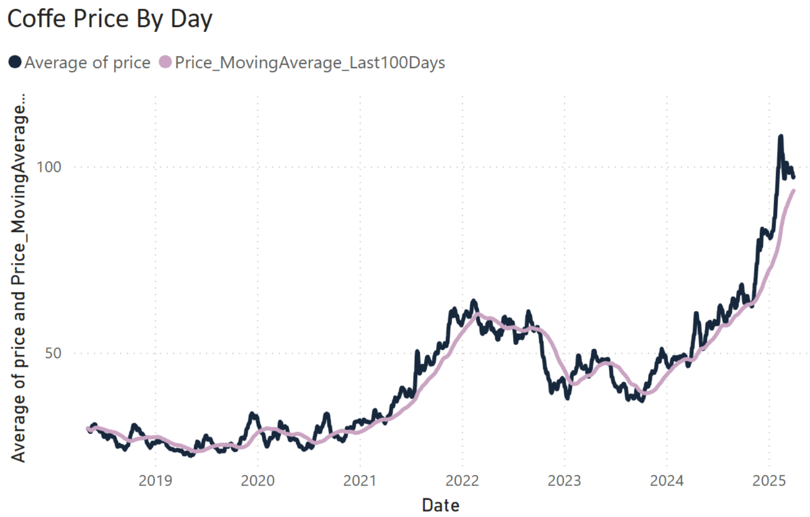
Calculating the Correlation Cofficient in Power BI with Dax

```
COR_Quantity_DaysLate =

    VAR MeanQuantity = AVERAGE(procurement_transactions[quantity])
    VAR MeanDaysLate = AVERAGE(procurement_transactions[Days Late])

    VAR SumNumerator =
        SUMX(
            procurement_transactions,
            (procurement_transactions[quantity] - MeanQuantity) * (procurement_transactions[Days Late] - MeanDaysLate)
        )

    VAR SumSqX =
        SUMX(
            procurement_transactions,
            (procurement_transactions[quantity] - MeanQuantity) ^ 2
        )

    VAR SumSqY =
        SUMX(
            procurement_transactions,
            (procurement_transactions[Days Late] - MeanDaysLate) ^ 2
        )

    RETURN
        DIVIDE(SumNumerator, SQRT(SumSqX * SumSqY))
```

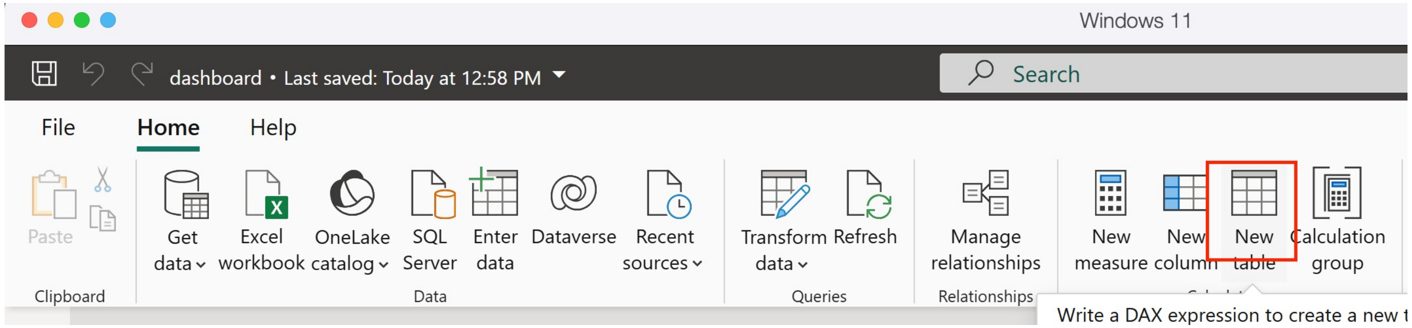
TIME BASED MEASURES

Calculating Moving Averages



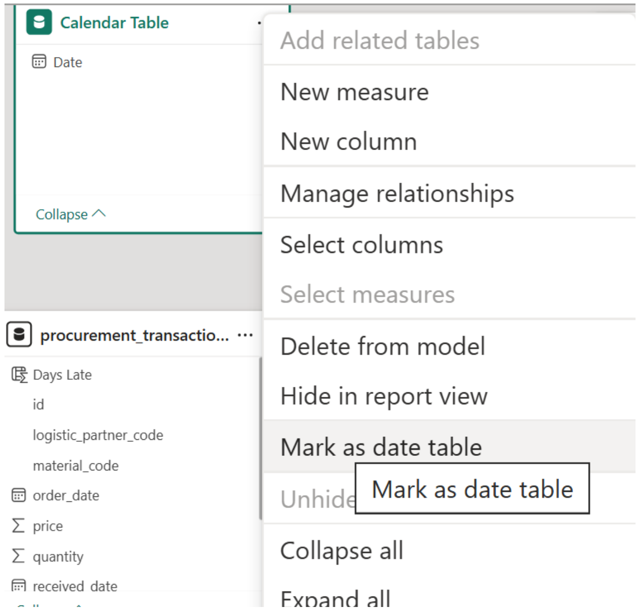


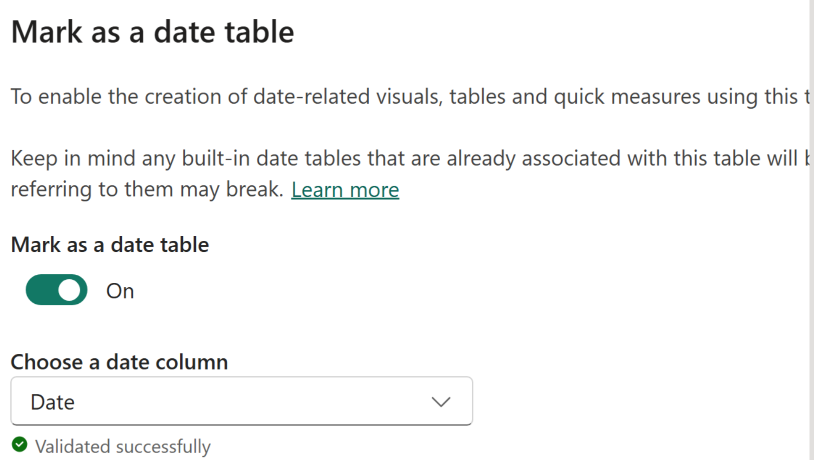
Create a Calendar Table

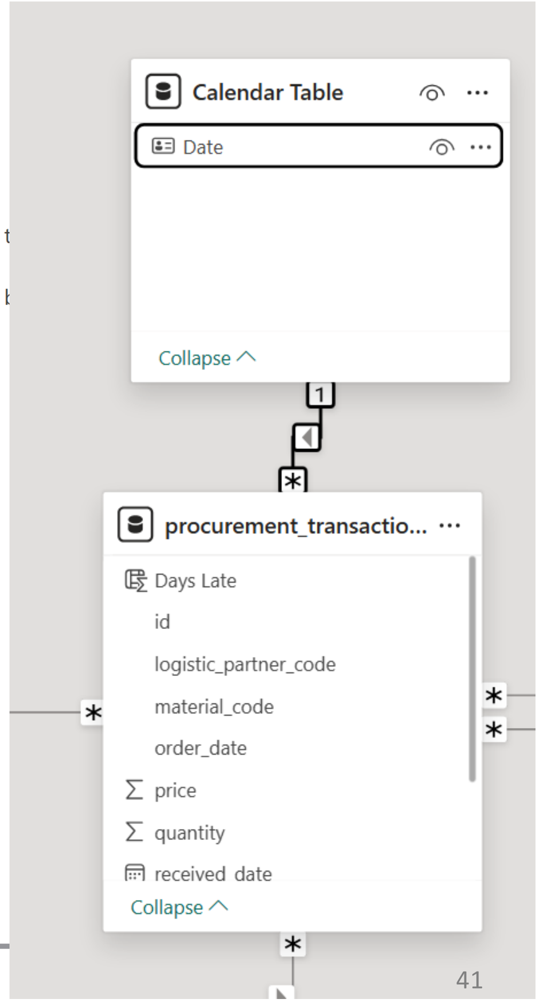



```
Calendar Table = CALENDAR(
        MIN(procurement_transactions[order_date]),
        MAX(procurement_transactions[order_date])
)
```

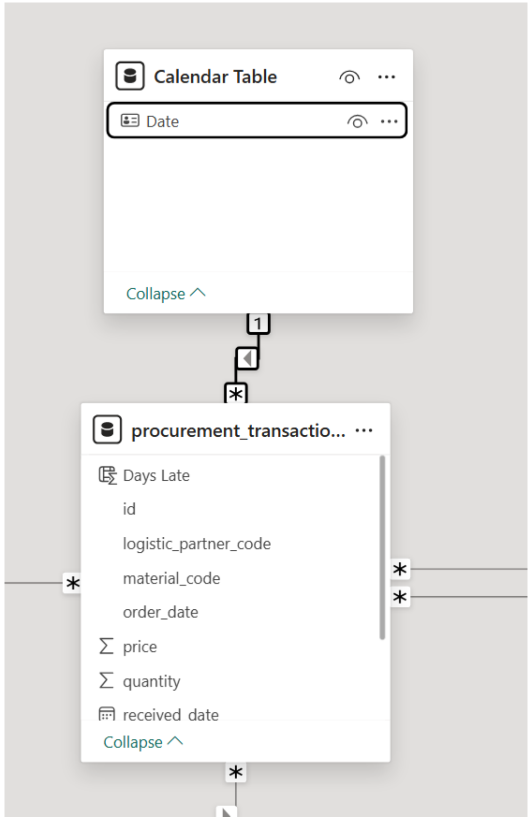
Create a Calendar Table







Connect Calendar Table to Procurement Transactions via Order Date

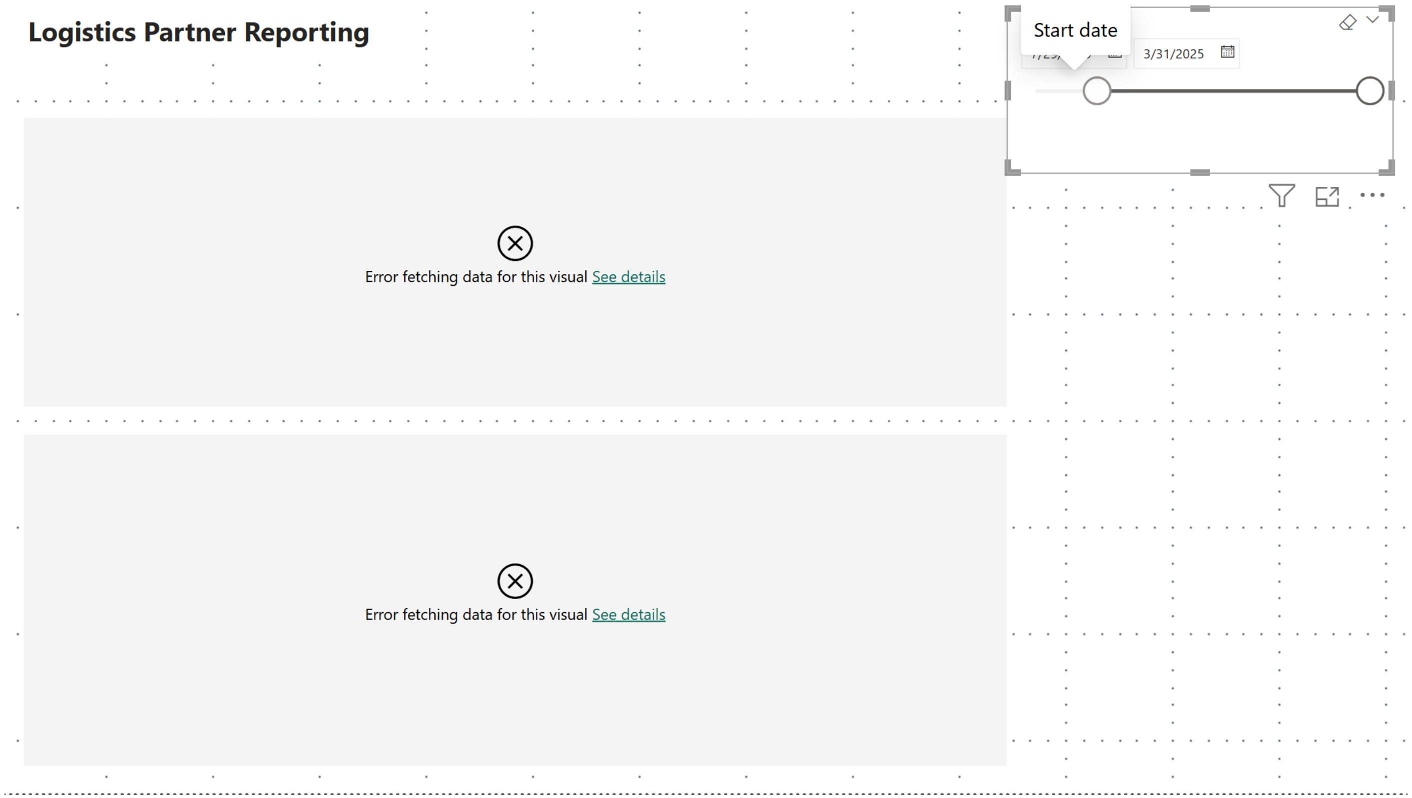


Add Measure to Procurement Table

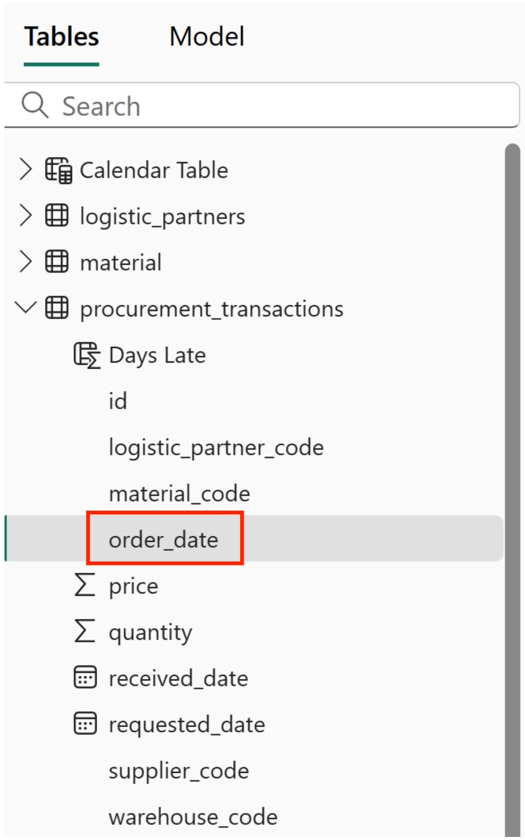
```
Price_MovingAverage_Last100Days = CALCULATE(
    AVERAGE(procurement_transactions[price]),
    DATESBETWEEN('Calendar Table'[Date],
    DATEADD(LASTDATE('Calendar Table'[Date]),-100,DAY),
    LASTDATE('Calendar Table'[Date]
)))
```

POTENTIAL ISSUES WITH DATE SLICERS

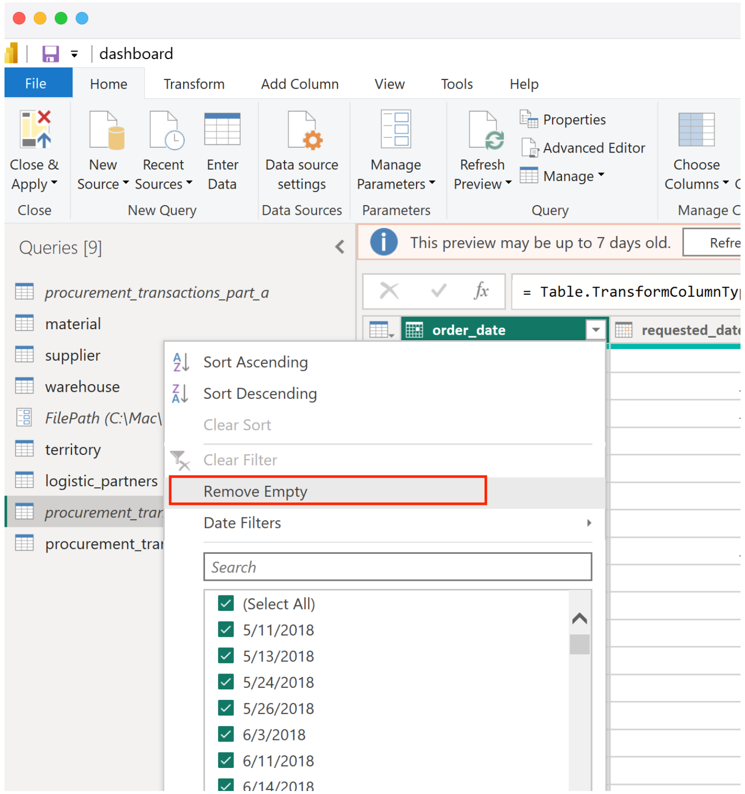
Issues with Date Slicers



Verify Column is Correctly Formatted / Recognized as Date



Potential Issue: Blank Values



Changing the Format How Data is Displayed in Visuals